Wednesday August 27

New set of valve design parameters:

				Efficiency (%)
(12)	4.78	2.60	1.83	3.6
(13)	2.54	1.00	1.00	57.0
(14)	0.57	1.21	3.68	49.3

Theorem 1.34 Suppose f is convex and continuously differentiable, then x^* is a global minimizer of f if and only if $\nabla f(x^*) = 0$. $(f: \mathbb{R}^n \to \mathbb{R})$

Proof: Let f be convex and continuously differentiable. Then $f(y) \geq f(x) + (y-x)^T \nabla f(x)$ for all $x, y \in \mathbb{R}^n$.

($\Leftarrow$) Suppose $\nabla f(x^*) = 0$. Then $f(y) \geq f(x^*) + (y-x^*)^T \nabla f(x^*) = f(x^*)$ for all $y \in \mathbb{R}^n$. That is, x^* is a global minimizer of f.

($\Rightarrow$) Suppose x^* is a global minimizer of f. Then $f(y) \geq f(x^*)$ for all $y \in \mathbb{R}^n$. Consider $y = x^* - \nabla f(x^*)$. We have

$$f(y) \geq f(x^*) + (y - x^*)^T \nabla f(x^*) = f(x^*) - \|\nabla f(x^*)\|^2.$$

for $f(y) \geq f(x^*)$ to hold for all y (and f continuous)

$\|\nabla f(x^*)\| = 0$ or equivalently, $\nabla f(x^*) = 0$. $\blacksquare$

Lemma 1.35 Function $f: \mathbb{R}^n \to \mathbb{R}$ is coercive if and only if every sublevel set of f is bounded.

Proof: Consider function $f: \mathbb{R}^n \to \mathbb{R}$. We show that f admits an unbounded level set if and only if f is not coercive.

C_z unbounded

$\Updownarrow$

$\exists \{x_k\} \ni \|x_k\| \to \infty$ with $f(x_k) \le z \quad \forall k$

$\Updownarrow$

f is not coercive. $\blacksquare$

Example: Given: $y_1 \cdots y_m \in \mathbb{R}^n$

$a_1 \cdots a_m \in \mathbb{R} \quad a > 0$

$\beta \in \mathbb{R}, \ \beta > 0$ "small"

$$f(x) = \underline{\left\| x - \frac{1}{m}\sum_{i=1}^{m} y_i \right\|} + \underline{\sum_{i=1}^{m} \frac{a_i}{(\|x - y_i\|^2 + \beta^2)^{1/2}}}$$

consider the problem $\min_{x \in \mathbb{R}^n} f(x)$

First term is minimized when x is the mean of the y's. It diverges as $\|x\| \to \infty$ in any way and is therefore coercive. It is also continuous. We could square the norm to make it smooth.

The second term is large whenever x is close to any y_i. Each term has max value a_i/β. As $\|x\| \to \infty$ in any way, this term decays to zero.

Overall this function is continuous, coercive and intuitively a global min exists ($f(x)$ has lower bound zero).

Theorem 1.37 If $f: D \to \mathbb{R}$ is continuous and D is closed and bounded, then a global minimizer (and a global maximizer) exists on D. (EXTREME VALUE THEOREM)

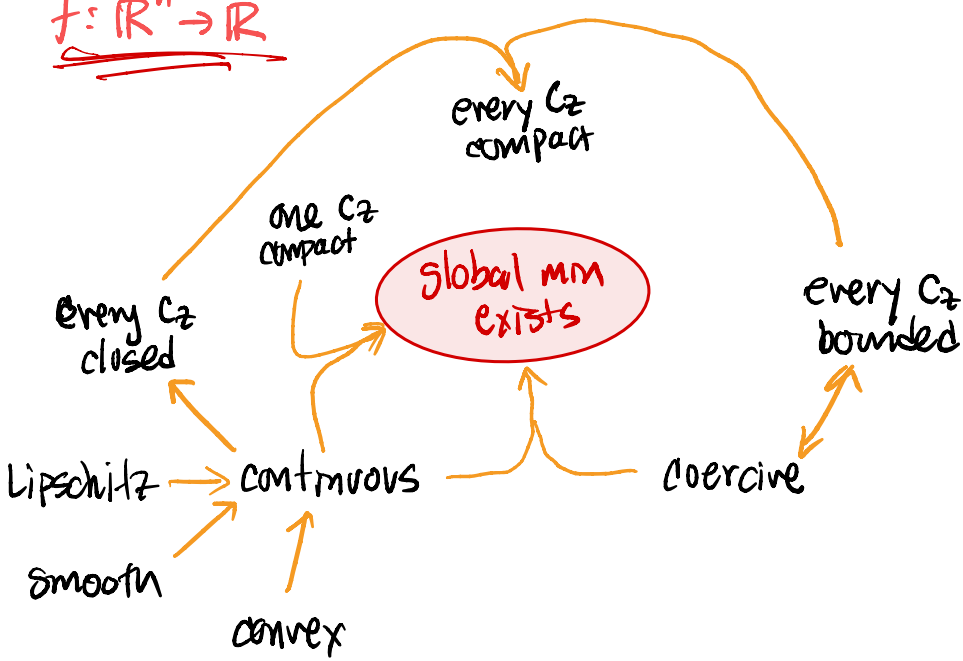
Theorem 1.36 If $f: \mathbb{R}^n \to \mathbb{R}$ is coercive and continuous then a global minimizer of f exists.

Proof: Consider coercive continuous function $f: \mathbb{R}^n \to \mathbb{R}$. Consider any nonempty sublevel set C_z. C_z is closed (continuity) and bounded (coercivity). So

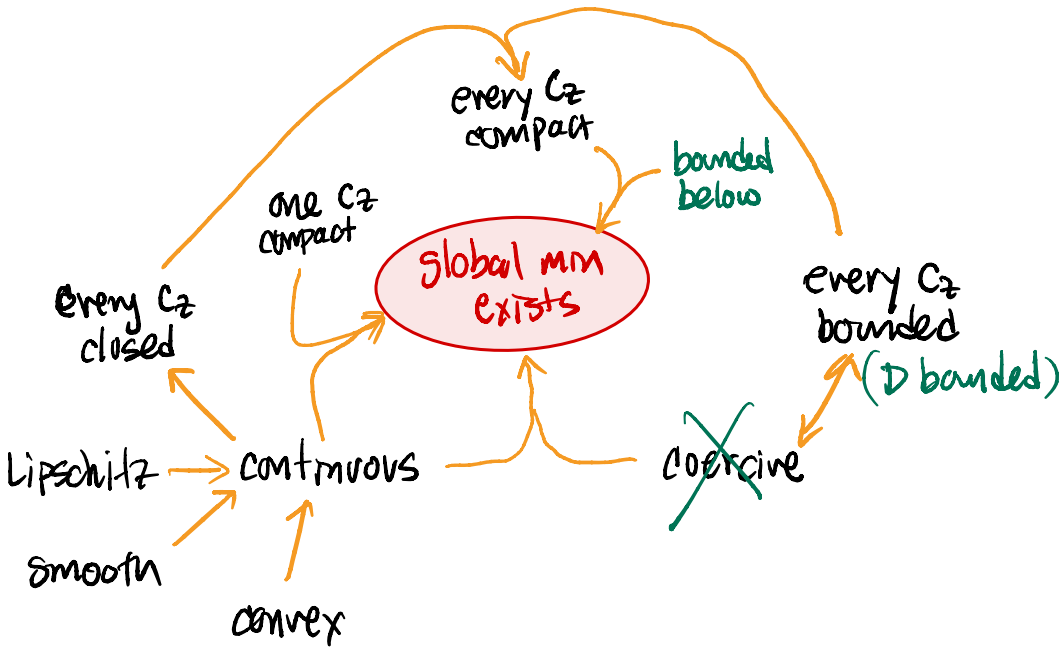
$$\underset{x \in \mathbb{R}^n}{\operatorname{argmin}} f(x) = \underset{x \in C_z}{\operatorname{argmin}} f(x)$$

By Theorem 1.37 a global min exists in C_z. $\square$

$f: \mathbb{R}^n \to \mathbb{R}$



$f: D \to \mathbb{R}$ $\quad D \subset \mathbb{R}^n$ $\quad D$ closed



Some thought challenges:

(1) Give an example of a function $f: \mathbb{R} \to \mathbb{R}$ which is not continuous but every sublevel set is closed.

(2) Give an example of a function $f: \mathbb{R} \to \mathbb{R}$ which is bounded below and continuous for which no global min exists.

(3) Give an example of a function $f: \mathbb{R} \to \mathbb{R}$ which is bounded below and coercive for which no global min exists.